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Diabetic gait and posture abnormalities: A biomechanical investigation through three dimensional gait analysis

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ABSTRACT

Background: Diabetic sensorimotor polyneuropathy is a long-term diabetic complication. It is involved in the pathogenesis of the diabetic foot, which is a major cause of morbidity and mortality. The study aims to investigate the effects of diabetic polyneuropathy on gait and posture.
Methods: Sixty seven subjects were enrolled: 21 diabetics without polyneuropathy, 26 with polyneuropathy, 20 controls (respectively, mean age 63.8 (SD 5.4), 63.2 (SD 5.6), 59.0 (SD 5.2) years, mean body mass index 26.3 (SD 2.5), 25.6 (SD 3), 24.0 (SD 2.9)). Postural and morphological evaluation and gait analysis were performed. Physical examination, together with a motion capture system synchronized with two force plates and two baropodometric systems were used. We evaluated lower limb mobility, foot deformities, trunk and pelvic posture, knee and heel position, plantar foot arch, three dimensional kinematics and kinetics during gait. The effect of peripheral vascular disease and microangiopathy on trunk and lower limb motion was also evaluated.
Findings: Trunk and lower limb joint mobility (in static and dynamic states) were more reduced in diabetics either with or without polyneuropathy on each plane; however in diabetics with polyneuropathy significantly lower ranges of motion were registered. Furthermore, both groups showed significant reductions in each joint moment and velocity ($P \leqslant 0.003$) during gait. In presence of both vasculopathy and microangiopathy a further significant reduction ($P \leqslant 0.001$) was noticed.
Interpretation: Altered gait and posture were found in diabetic patients irrespective of polyneuropathy. This approach may be relevant to predict the risk ulceration before clinically detectable neuropathy.
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1. Introduction

Diabetic sensorimotor polyneuropathy (DSP) is the most common long-term diabetic complication, and it is involved in the pathogenesis of the diabetic foot (Perkins and Bril, 2003). The prevalence of DSP in diabetic patients is 25% after 10 years of disease, and it is often associated with peripheral arterial disease, nephropathy and retinopathy.

DSP affects sensory, motor and autonomic nerves, and leads to progressive degeneration and nerve fiber loss (Abbott et al., 1998; American Diabetes Association, 1988; Andersen et al., 1997; Perkins and Bril, 2003). The deleterious effects of DSP include vibration, tactile and perception sensory loss in the large fibers, and a lack of thermal and pain sensitivity in the small fibers.

This loss of sensitivity increases the risk for developing ulcers (Andersen and Jakobsen, 1999; Van Schie, 2005). Furthermore the lack of somatosensory information from the feet and legs for control, and postural stability leads to the increase in postural sway in stance (Horak et al., 2002). It has been demonstrated that DSP subjects showed significantly more postural sway than control subjects (CS) both with eyes open and eyes closed (Simoneau et al., 1995; Dickstein et al., 2001). Nevertheless motor neuropathy leads to structural changes in the foot, characterized by weakness of the intrinsic lumbrical and interosseal muscles; it also induces progressive atrophy and consequent toe deformities. Both foot deformities and limited joint mobility appear to be strong predictors of abnormal mechanical foot loading during walking since they can result in excessive plantar loads and gait alterations (Andersen et al., 1997; Bevans, 1992). In addition to neuropathy and vascular disease, trauma is needed to cause tissue breakdown (Van Schie, 2005). Thus, the study of foot biomechanics is of major importance in ulcer prevention. One of the principal functions of the foot is its shock-absorbing capability during heel strike and its adaptation to the uneven surface of the ground during gait (Van Schie, 2005). For

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instance, several authors investigated DSP subjects' gait. Courtemanche et al., (1996) and Mueller et al., (1994) found in DSP subjects lower walking velocity and stride length.

Dingwell et al. (1999,2000) and Dingwell and Cavanagh (2001) investigated the variability of the sagittal plane kinematics of DSP, non-neuropathic (NoDSP) and control subjects during motorized treadmill walking. A trend toward increased variability within the three groups was found for several measures of gait variability, even though most of them were not significant. Katoulis et al., (1997) found slower walking speed, smaller peak of both knee joint angle in the sagittal plane and ankle joint moment in the frontal plane for DSP with previous history of ulceration. Also a higher peak value of the vertical and anteroposterior ground reaction forces was found in those subjects. Several studies assessed both plantar pressure and ground reaction forces in order to understand the aetiology of diabetic foot (Cavanagh et al., 1993, 2000; Donovan et al., 2008; Giacomozzi et al., 2002; Shaw et al., 1998; Uccioli et al., 2001). Although these studies showed alterations in some gait parameters, no studies reported full-body three dimensional gait analysis of both DSP and NoDSP subjects.

Therefore, the aim of this study was to develop a methodology which identifies subjects at risk for ulceration based on both diabetic gait and postural alterations, through three dimensional gait analysis. Our approach was based on the simultaneous acquisition of motion analysis, ground reaction forces, and plantar pressure data along with physical and postural examinations.

Furthermore the effect of peripheral vascular disease and microangiopathy (microalbuminuria and retinopathy) on trunk and lower limb motion was also evaluated, as they are considered major risk factors for plantar ulcers in diabetic foot (Abbott et al., 1998; Van Schie, 2005).

2. Methods

2.1. Patient recruitment

Diabetic patients and CS were recruited from the patients attending the outpatient clinic of the Department of Metabolic Disease at the University of Padova (Italy), as well as from university personnel. Inclusion criteria were: either type 1 or type 2 diabetic subjects with walking ability, no history of ulcers or neurological disorders (apart from DSP), and no history of orthopedic problems, lower limb surgery, or cardiovascular disease.

On the basis of these criteria, 67 patients were enrolled: 20 CS (mean age 59.0 (SD 5.2) years, mean BMI 24.0 (SD 2.9)), 21 NoDSP subjects (mean age 63.8 (SD 5.4) years, mean BMI 26.3 (SD 2.5)), and 26 patients with DSP (mean age 63.2 (SD 5.6) years, mean BMI 25.6 (SD 3)). All subjects gave a written informed consent. The protocol was approved by the local Ethics Committee.

The subjects' feet were checked for skin lesions, bone deformities, ulcerations, signs of infection and previous amputations. Height (m) and weight (wearing only undergarments, without shoes) were recorded and body mass index (BMI) (kg/m^2) was calculated.

The neurological evaluation included the assessment of symptoms and signs compatible with peripheral nerve dysfunction. The Michigan Neuropathy Screening Instrument questionnaire, which evaluates motor and sensory symptoms, was filled out (Feldman et al., 1994). Subjects were classified as neuropathic if they were found to be positive for three or more out of a total of fifteen specified symptoms (American Diabetes Association, 1988). The physical examination consisted of: 1. patellar and ankle reflexes, with the patient in the sitting position; 2. assessment of muscle strength by ability to walk on heels, bilateral dorsiflexion/plantar flexion of the feet, flexion/extension of legs, abduction/adduction of both forearms and fingers, all against resistance; 3. sensory testing carried out on the index finger, and on the hallux (pin-prick with a disposable 25 mm/7 mm needle), touch (10 g Semmens Weinstein monofilament, pathologic if no response on 3 out of 10 sites) and vibration perception threshold (VPT, 128 MHz tuning fork and Biothesiometer, pathologic if >25 V); 4. pain sensitivity; 5. electroneurophysiological (EMG) study; 6. ankle-to-brachial systolic pressure ratio (Index of Winsor). Cardiovascular autonomic tests were also performed: if two or more tests were abnormal the patient was considered to have autonomic neuropathy.

HbA1c values from the preceding 10 years were collected. Each patient had at least one ophthalmologic examination, a urinary albumin-to-creatinine ratio measured (0–30 mg/g normal, 30–300 mg/g microalbuminuria, >300 mg/g macroalbuminuria), a carotid artery Doppler ultrasound examination, and a 12-leads electrocardiogram in the three month period preceding the study.

2.2. Postural and physical examination

Subjects underwent a morphological examination of the foot to assess the type of foot (cavus, planus, normal), foot deformities (hallux valgus/normal/rigidus, claw and hammer toes, limitation of dorsiflexion of the great toe, fifth abducted/adducted toe, overlapping toes), and pre-/post-surgery ulcer lesions (bunions, calluses, scars, soft corns). Hip, knee, and ankle joint mobility was assessed. The magnitude of the sagittal curvature was measured: considering the pelvic structure on the sagittal plane, either a forward or a backward displacement of the point tangential to the sacrococcygeal plane was estimated. Based on these measurements, an anteverse pelvic position or a posterior pelvic tilt in combination with a forward or a backward displacement of the pelvis was evaluated. Measurements were also made on the functionality of the pelvic cingulum (pelvic translation, pelvic rotation, pelvic bascule), cingulum scapular rotation (rotation of the clavicular acromion with respect to the contralateral side), and cingulum scapular flexion (fall of the clavicular acromion with respect to the contralateral side). Any leg length asymmetry greater than 5 mm was taken into account and heel position and plantar foot arch during bipedal load were also assessed (Bourdiol, 1980). Both a footprint (Bourdiol, 1980) and a static acquisition (Kapteyn et al., 1983) on the plantar pressure system were used.

2.3. Motion analysis

Motion analysis was performed with a BTS motion capture system (six cameras, 60–120 Hz) synchronized with two Bertec force plates (FP4060–10), and integrated with two Imago plantar pressure systems (0.64 cm^2 resolution, 150 Hz). A full-body marker set was used (Leardini et al., 2007; Sawacha et al., 2005). The center of the femoral head was assumed to coincide with the center of the acetabulum: the latter was reconstructed by means of a functional method proposed by Cappozzo et al. (1995). The positions of most of the anatomical landmarks were assumed to coincide with the center of the marker used. Lateral and medial epicondyle, medial malleolus, and great trochanter positions were calibrated using static acquisition without the aid of a pointer (Cappozzo et al., 1995). Four extra markers were applied on each thigh in order to create two clusters which were used to calibrate the position of great trochanter, lateral and medial epicondyles.

2.4. Static acquisition

In the present study the static acquisition was performed with a full-body marker set: the subject was asked to stand on the compound system comprising both the plantar pressure and dynamo-

metric platforms without shoes, with the heels jointed, and with the feet 30° apart (measured between the two medial parts of the feet) (Kapteyn et al., 1983; Sawacha et al., 2009).

2.5. Dynamic acquisition

Subjects were asked to walk along the walkway so that the target foot would naturally land on the force plates. Three walking trials, at a self-selected speed, with three right and three left foot contacts on both the force and pressure plates were carried out. Velocity, stride and step parameters were calculated. The patients' kinematic movements were recorded utilizing reflective markers. Kinetic data were calculated using the force generated in the stance phase, which was recorded with two Bertec straingauged force plates. For each trial, angular displacements and internal joint moments were plotted over one gait cycle. The displacement of the center of pressure pattern was analyzed and both the mean and range of the mediolateral and anteroposterior excursions evaluated (Giacomozzi et al., 2002). Based on this methodology each footprint was compared with the others after rotating each one according to its longitudinal axes. The latter was defined as the line connecting the projection of the 2nd metatarsal head and the calcaneus markers on the footprint. So far the lateral side of the foot (on the right side of the longitudinal axis of a right footprint and on the left side of the longitudinal axis of a left footprint) was considered as the positive mediolateral direction, and the medial side of the foot as the negative one.

2.6. Statistical analysis

The sample size was calculated with Altman Nomogram (Altman, 1991).

The evaluation of the confidence intervals for the observed proportions and the correlation analysis (Pearson) were performed, respectively, with the staRt Package and the RODBC Package of R statistical software. Multiple regression analysis (peripheral vascular disease and microangiopathy as independent variables, postural and morphological evaluation related variables as dependent variables) was performed with SPSS software.

A *t*-test (Matlab software) was used to compare the following variables (between paired groups of subjects: DSP vs. NoDSP, DSP vs. C and NoDSP vs. C): age, duration of the disease, HbA1c, BMI, time and space parameters, center of pressure measurements. Regarding joint rotations and moments, each subject's variables were represented by the mean from three representative walking trials. Intra-class correlation was used to aid in selecting which of each subject's representative walking trials were to be included in the computation of the mean; thus the intra-class correlation coefficient was calculated for each subject's kinematic and kinetic parameters. Walking trials with an intra-class correlation coefficient less than 0.75 (75%) were excluded from the statistical analysis (Maynard et al., 2003).

One-way ANOVAs were performed for the joint rotation angles and moments in order to identify significantly different variables. Comparisons were made for joint rotations between the NoDSP and CS groups and between the DSP and CS groups. The results of the statistical analysis were expressed relative to specific phases of the gait cycle (Perry, 1992), where significant differences ($P < 0.05$) were found between groups for joint rotations. Comparisons were also made for joint rotation moments between the NoDSP and CS groups and between the DSP and CS groups. The results of this statistical analysis were expressed relative to specific phases of the stance phase of gait, where significant differences ($P < 0.05$) were found between groups for joint rotation moments.

3. Results

The clinical characteristics of the study subjects are reported in Table 1.

As shown in Table 1, all patients were under fair metabolic control (Takeshi Oomichi et al., 2006). The DSP group had a higher prevalence of both micro- and macrovascular complications.

Table 1
Clinical characteristics, center of pressure (COP) and time and space parameters (mean and standard deviation (SD)) of diabetic neuropathic group (DSP), diabetic non-neuropathic group (NODSP), control group (CS). The reported *P* values indicate the results of the comparison between the DSP and NoDSP groups, the DSP and CS groups, and the NoDSP and CS groups. A value of $P < 0.05$ was considered statistically significant (P^*).

	NoDSP		DSP		CS		DSP vs. NoDSP *t*-test (*P*)		DSP vs. C *t*-test (*P*)		NoDSP vs. C *t*-test (*P*)	
Subjects (%)	30.3		39.4		30.3							
Age (years)	63.8(5.4)		63.2(6.0)		59.0(2.9)		0.70		0.004*		0.01*	
BMI	26.3(2.0)		25.6(2.9)		24.0(2.8)		0.4		0.2		0.02*	
Years of disease (years)	17.2(11.7)		22.1(14.3)				0.2					
Sex (% of subjects)	F	M	F	M	F	M	F	M	F	M	F	M
	30	70	42.3	57.7	35	65	0.19	0.07	0.31	0.77	0.17	0.24
Diabetic retinopathy (% of subjects)	40.0		65.4				0.04*					
Microalbuminuria (%of subjects)	25.0		34.6				0.1					
Peripheral vascular disease (%of subjects)	15.0		19.2				0.6					
Autonomic neuropathy (% of subjects)	0		23.0				0.02*					
Type I (% of subjects)	15.0		42.3				0.02*					
Type II (% of subjects)	85.0		57.7				0.02*					
Hb A_{1c} (%)	7.7(1.1)		7.6(1.7)				0.1					
Gait velocity (m/s)	1.10(0.2)		1.10(0.2)		1.27(0.1)		0.9		0.003*		0.003*	
Stride period (s)	1.13(0.1)		1.19(0.1)		1.1(0.1)		0.3		0.3		0.4	
Stride length (m)	1.23(0.1)		1.20(0.2)		1.4(0.1)		1.0		0.000*		0.002*	
Stance period (s)	0.66(0.1)		0.73(0.1)		0.5(0.1)		0.1		0.000*		0.000*	
COP range of excursion AP (cm)	23.6(3.3)		22.4(4.4)		24.6(2.4)		0.3		0.0001*		0.07	
COP range of excursion ML (cm)	1.7(0.5)		1.9(0.7)		2.0(0.6)		0.4		0.5		0.003*	
COP mean excursion ML (cm)	0.18(0.8)		−0.20(1.4)		0.30(0.9)		0.3		0.04*		0.5	
COP mean excursion AP (cm)	14.6(2.6)		14.1(3.8)		15.3(2.1)		0.6		0.06		0.2	

In analyzing the data from the postural and physical examinations, hip joint mobility was found to be significantly reduced in the NoDSP ($P \leqslant 0.047$) as well as in the DSP group ($P \leqslant 0.056$) when compared with the CS group. Knee recurvatum was more prevalent in the NoDSP group ($P \leqslant 0.012$) and this prevalence was found to be statistically significant compared to the CS group.

The evaluation of foot arch morphology showed a greater prevalence of cavus foot both in the DSP ($P \leqslant 0.056$) and the NoDSP ($P \leqslant 0.046$) group. A higher percentage of subjects with toe deformities was found both in the DSP (hammer toes $P \leqslant 0.053$, claw toes $P \leqslant 0.071$) and NoDSP (hammer toes $P \leqslant 0.012$) group when compared with the CS group. Plantar callosity was more prevalent in the DSP ($P \leqslant 0.001$) and NoDSP groups ($P \leqslant 0.034$) than in the CS group. *Z*-test analysis was also performed on the DSP group to account for the relative contributions of retinopathy, microalbuminuria, and peripheral vascular disease to biomechanical alterations. The presence of these complications was associated with more serious postural alterations. The scapular cingulum flexion was significantly decreased in the NoDSP subjects in association with microalbuminuria ($P \leqslant 0.003$), retinopathy ($P < 0.006$), and peripheral vascular disease ($P \leqslant 0.008$). Moreover a remarkable correlation was found between this reduction in the scapular cingulum motion and the presence of microalbuminuria ($R = 0.6$), retinopathy ($R = 0.7$), and peripheral vascular disease ($R = 0.8$). A significant reduction in pelvic translation was also observed in association with retinopathy when compared with NoDSP subjects without complications ($P \leqslant 0.03$). A significantly greater prevalence of toe deformities was observed in association with microalbuminuria ($P \leqslant 0.0004$, $R = 0.8$), retinopathy ($P \leqslant 0.002$, $R = 0.2$) and peripheral vascular disease ($P \leqslant 0.002$, $R = 0.4$). The contribution of the combination of peripheral vascular disease and microangiopathy to predicting postural and morphological alteration was reported in Table 3. The center of pressure displacement showed significant reductions in the mediolateral and anteroposterior directions in both the DSP and NoDSP subjects (see Table 1), when the baropodometric data of walking trials were examined.

3.1. Motion analysis

In performing the analysis of the gait parameters the data of 66 subjects were analysed (one subject refused the test).

Joint rotation and moment normative bands (the mean plus and minus one standard deviation) were created using the CS group's data. The data from the other groups were compared to these.

In the report on joint rotations, abduction/adduction and internal/external rotation of the knee were not considered although the model accounts for their values because these joint rotations were not shown to be feasible when reconstructed through markers placed directly on the skin (Della Croce et al., 2005).

The joint rotation ranges of motion from the DSP and NoDSP groups were compared with those from the CS group and the results of this comparison are reported in Fig. 1.

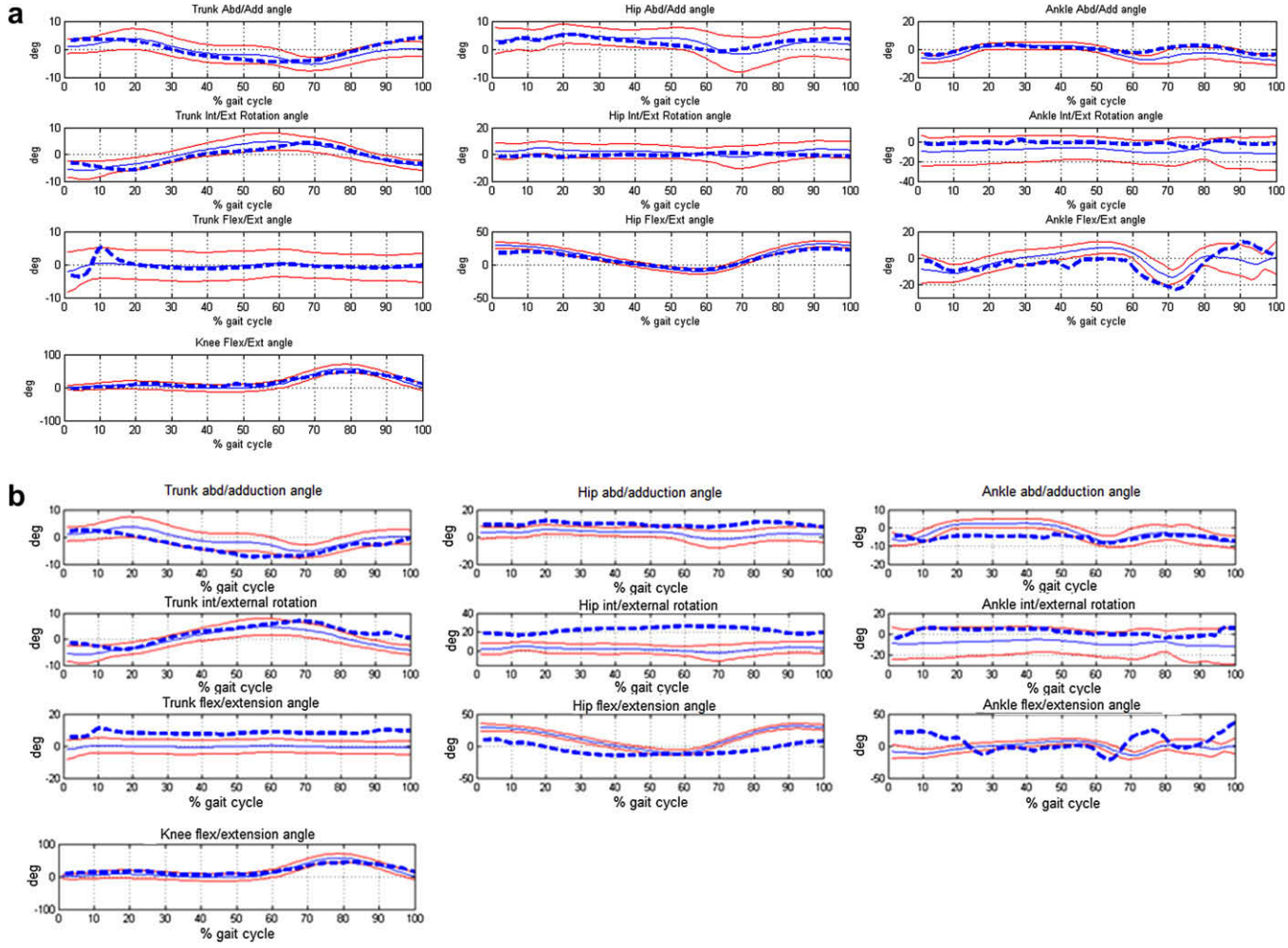


Fig. 1. Gait analysis joint kinematic results. Mean joint angles of: (a) NoDSP and (b) DSP, over the normative bands (CS) are reported. Horizontal axis depicts one gait cycle; vertical axis represents (a–b) angles degrees.

Table 2
Mean, standard deviation (SD) and ANOVA results for joint rotation angles computed on the diabetic neuropathic group (DSP), diabetic non-neuropathic group (NODSP), control group (CS) over the gait cycle. A value of $P < 0.05$ was considered statistically significant: NS = nonsignificant, P^{*} indicate results of the comparison between the DSP and CS groups, P^{**} indicate results of the comparison between NoDSP and CS groups. T = trunk, H = hip, K = knee, A = ankle. 1 = abduction–adduction, 2 = internal–external rotation, 3 = flexion–extension. IC = initial contact, LR = loading response, MS = mid-stance, TS = terminal stance, PS = pre swing, IS = initial swing, MSW = mid swing, TSW = terminal swing.

	Rotation (°) mean (SD)			IC		LR		MS		TS		PS		IS		MSW		TSW	
	NODSP	DSP	CS	P^{*}	P^{**}	P^{*}	P^{**}	P^{*}	P^{**}	P^{*}	P^{**}	P^{*}	P^{**}	P^{*}	P^{**}	P^{*}	P^{**}	P^{*}	P^{**}
T1	−1.2(6)	1.4(3)	0.4(3)	0.1n.s	2.0n.s	0.1n.s	0.9n.s.	0.04	0.001	0.04	0.04	0.04	0.003	2.0n.s	0.8n.s	0.1n.s	1.0n.s	0.4 n.s	2.4 n.s
T2	−1.9(3)	0.1(3)	1.3(4.)	0.04	0.04	0.01	0.02	0.05n.s	0.04	0.3n.s	0.003	0.05n.s	0.01	0.03	0.4 n.s	0.001	1.4n.s	0.001	0.1n.s
T3	−1.4(0.1)	3.4(0.6)	2.5(1)	0.04	0.1n.s.	0.04	1.3n.s.	0.04	0.1n.s	0.04	0.1n.s	0.04	0.05n.s	0.04	0.05n.s	0.04	0.05n.s	0.04	0.05n.s
H1	3.7(4)	−4.7(9)	8.8(4)	0.03	0.02	0.03	0.001	1.5	0.003	0.2n.s.	0.1n.s.	0.4n.s	0.6n.s.	0.004	0.01	0.01	0.04	0.04	0.04
H2	−0.0(0.4)	2.7(2)	10.7(2.9)	0.001	0.001	0.001	0.02	0.004	0.04	0.004	0.04	1.0n.s	0.5n.s	0.04	0.02	0.005	0.002	0.0001	0.002
H3	−0.8(1)	4.6(12)	15.9(18)	0.1n.s	0.02	0.1n.s	0.04	0.05n.s	0.04	0.05n.s	1.0n.s.	0.05n.s	0.01	0.05n.s	0.8n.s	0.05n.s	0.004	0.05n.s	0.04
K3	9.3(24)	13.5(19)	23.1(20)	0.04	0.2n.s.	0.04	0.1n.s.	0.04	0.2n.s.	0.04	0.001	0.04	0.03	3.9n.s	0.4n.s.	0.01	0.02	0.04	0.01
A1	−6.7(2)	−1.2(6)	2.4(5)	0.2n.s	0.9n.s	0.02	0.04	0.003	0.01	0.003	0.005	0.1n.s	0.05n.s	0.3n.s	1.1n.s	0.05n.s	1.7n.s	0.05n.s	1.3 n.s
A2	0.0(2)	−4.4(1)	1.1(1)	0.3n.s	0.001	2.5n.s	0.02	0.002	0.01	0.01	0.004	0.01	0.01	0.01	0.04	0.8n.s	0.04	0.04	0.04
A3	−6.0(7)	−0.5(5)	2.5(8)	0.03	0.003	0.03	0.1n.s.	0.03	0.05n.s.	0.3n.s	0.02	0.3n.s	0.04	0.01	0.04	0.1n.s	0.02	0.01	0.02

One-way ANOVAs were carried out between the DSP and CS groups, and between the NoDSP and CS groups, with the results about the kinematics variables reported in Table 2. When considering the joint moments, significant differences were observed on each joint over the entire stance phase of gait ($P < 0.04$) except for: the trunk internal-external rotation and the knee flexion–extension during the initial contact ($P \geqslant 0.05$ in NoDSP and DSP); the hip abduction–adduction between the loading response and the pre swing ($P \geqslant 0.1$ in DSP), the ankle abduction–adduction during the first 10% of the stance phase ($P \geqslant 0.2$ in DSP). A one-way ANOVA was also performed on the NoDSP group taking into account the roles of retinopathy, microalbuminuria, and peripheral vascular disease in determining gait pattern alterations. The results of this analysis showed that subjects with these complications have statistically significant differences in joint kinematic and kinetic parameters over a larger part of the gait cycle than those in the same group without complications, and this is true for most of the parameters measured. In the presence of microalbuminuria, trunk kinematics showed statistically significant differences over the entire gait cycle for the internal-external rotation angle. In the presence of peripheral vascular disease, the following kinematic parameters showed statistically significant differences over various phases of the gait cycle, with their respective phases of the gait cycle reported: trunk flexion–extension angle (full gait cycle), hip abduction–adduction (initial contact, loading response, midstance, terminal stance, pre swing, initial swing, mid swing), knee flexion–extension angle (full gait cycle), ankle joint flexion–extension (full gait cycle), and abduction–adduction (full gait cycle). In the presence of microalbuminuria, the trunk internal-external rotation moment showed a statistically significant difference over the full stance phase compared to the NoDSP group without any complications.

In the time and space parameters' analysis (Table 1), DSP subjects displayed significant differences almost for each variable when compared to CS subjects. DSP time and space parameters were also compared with those from the NoDSP group, but no significant differences were observed. When the presence of retinopathy, peripheral vascular disease or microalbuminuria was considered, patients with peripheral vascular disease or microalbuminuria in the NoDSP group showed statistically lower stride length ($P \leqslant 0.001$) than patients without these complications.

4. Discussion

Our approach shows marked alterations in the motion patterns of DSP subjects: these findings are in agreement with but also significantly extend those of Courtemanche et al. (1996), Dingwell et al. (1999) and Katoulis et al. (1997), who demonstrate that neuropathy leads to disturbances in foot mechanics. We also showed that, in NoDSP subjects, there is an altered pattern of motion, as verified by the identification of early alterations in the gait of the diabetic subjects without a clinical diagnosis of neuropathy. In DSP patients, the reduction of joint moments, which becomes more pronounced when moving from the most proximal to the most distal joint, suggests that neuropathy causes progressive damage from the proximal to the distal limb. This condition imposes the need for a change in the walking strategy from the ankle to the hip joint, a situation known as *hip walking strategy* (Mueller et al., 1994).

Similar alterations were found for joint motion in the NoDSP group, even though differences between joint motion patterns in the NoDSP and CS groups were reduced when moving from the distal to the proximal limb. The knee joint showed a reduction in the flexion–extension range of motion in agreement with Katoulis' data. This could be associated with a compensatory mechanism to the ankle joint stiffening in order to keep normal postural balance; a possible association with tissue glycation at the knee joint cannot be excluded. The longer stance period observed in the DSP group is consistent with previous findings (Mueller et al., 1994) which reported this alteration as a typical consequence of neuropathic disease.

A larger presence (not significant: $P > 0.05$) of subjects with an anteverted position of the pelvis was observed in the DSP patients when compared with NoDSP ($P = 0.57$) and CS ($P = 0.28$) patients. This was unexpected considering the greater prevalence of cavus foot found in this group, which is generally associated with a backward displacement of the pelvis and a posterior pelvic tilt (Bourdiol, 1980). This finding might explain why the DSP subjects revealed a gait similar to a "flat footed gait" when analyzed with baropodometry and when the center of pressure displacement during gait was examined. This condition has been described previously by others, and it may be caused by plantar tissue thickening derived from tissue glycation in diabetics (D'Ambrogi et al., 2003; Giacomozzi et al., 2002; Mueller et al., 1994; Sinacore et al., 2008; Van Schie, 2005).

Reduced coxofemoral mobility was found in both the DSP and NoDSP groups when compared with the CS group: this could be interpreted as an early coxopathy.

In DSP subjects, no postural compensatory mechanisms for the anatomic dysmetria were observed: thus, when compared to NoDSP and CS subjects, lower cingulum scapular flexion and lower pelvic translation were observed. The lack of such a compensatory mechanism could be connected to a DSP proprioceptive deficit, which would affect balance control. This finding seems to be in agreement

Table 3
Prediction of postural and morphological alterations, diabetic group only. *a* = Analysis of variance. *b* = Pearson correlation.

Dependent variable	Independent variable	Unique contribution			Overall model	
		R	R^b	P^a	R^2	P^a
Cavus foot	Retinopathy	−0.3	−0.3	−0.3	0.2	0.2
	Microalbuminuria	0.1	0.4	−0.1		
	Peripheral vascular disease	0.2	0.2	0.2		
Planus foot	Retinopathy	0.06	−0.4	−0.5	0.4	0.02
	Microalbuminuria	−0.5	−0.3	0.3		
	Peripheral vascular disease	0.05	0.1	0.04		
Valgus heel	Retinopathy	−0.2	0.2	−0.1	0.1	0.4
	Microalbuminuria	−0.3	−0.05	0.6		
	Peripheral vascular disease	0.4	0.3	0.5		
Varus heel	retinopathy	0.05	−0.2	0.5	0.3	0.08
	microalbuminuria	0.4	0.2	−0.4		
	peripheral vascular disease	0.8	0.4	0.04		
Toes deformities	Retinopathy	−0.2	0.5	−0.2	0.3	0.09
	Microalbuminuria	0.2	0.8	0.4		
	Peripheral vascular disease	0.4	0.02	0.5		
Pelvic rotation	Retinopathy	0.3	0.01	0.06	0.1	0.5
	Microalbuminuria	0.1	0.2	0.3		
	Peripheral vascular disease	0.1	0.7	0.8		
Pelvic translation	Retinopathy	0.2	−0.2	−0.2	0.1	0.6
	Microalbuminuria	−0.03	0.02	0.03		
	Peripheral vascular disease	0.4	0.3	0.4		
Pelvic bascule	Retinopathy	−0.4	0.1	−0.005	0.2	0.4
	Microalbuminuria	0.1	0.5	−0.6		
	Peripheral vascular disease	0.1	0.7	0.9		
Pelvis anterversion	Retinopathy	−0.4	−0.05	−0.02	0.2	0.2
	Microalbuminuria	0.1	−0.3	0.2		
	Peripheral vascular disease	0.08	0.8	0.9		
Pelvis retroversion	Retinopathy	0.3	−0.4	0.1	0.2	0.4
	Microalbuminuria	0.3	0.3	0.6		
	Peripheral vascular disease	0.7	0.1	0.6		
Hip ipomobility	Retinopathy	0.3	0.07	0.09	0.07	0.7
	Microalbuminuria	0.6	0.6	0.6		
	Peripheral vascular disease	0.3	0.8	0.7		
Recurvatum knee	Retinopathy	−0.1	0.1	0.09	0.04	0.8
	Microalbuminuria	−0.4	−0.1	−0.4		
	Peripheral vascular disease	0.5	0.5	0.7		
Valgus knee	Retinopathy	0.2	0.04	−0.3	0.1	0.6
	microalbuminuria	−0.1	0.2	0.3		
	Peripheral vascular disease	0.3	0.9	0.2		
Cingulum scapular flexion	Retinopathy	0.07	0.2	0.3	0.2	0.4
	Microalbuminuria	0.1	0.7	−0.2		
	Peripheral vascular disease	0.8	0.4	0.2		
Leg length asymmetry	Retinopathy	0.1	0.05	0.2	0.1	0.5
	Microalbuminuria	0.1	0.2	0.2		
	Peripheral vascular disease	0.7	0.8	0.3		

with Courtemanche, who associated DSP with a reduced sensibility to touch and vibration and a diminished sense of limb position, as a result of the loss of large fiber proprioceptive and somatic feedback from receptors in the legs and feet and/or from muscular weakness due to motor neuropathy (Courtemanche et al., 1996).

One of the aims of this study was to investigate the biomechanics of the diabetic foot with respect both to postural and gait analysis. We also assessed the potential contribution of both micro- and macroangiopathic complications to the biomechanics of the diabetic foot. Remarkably, more severe alterations in both gait and posture parameters were found in the presence of peripheral vascular disease: this seems to suggests an important and additive role of vascular disease in gait performances. Of notice, severe biomechanical alterations were found among patients with microalbuminuria, which correlated with the greater prevalence of toe deformities observed in this group of subjects ($R = 0.8$). Further studies will be needed to determine the pathophysiological basis for this association. Our findings seem to be in contrast with Lavery et al. (1998) who previously evaluated several potential risk factors for diabetic foot ulcerations. In his report duration of diabetes mellitus of more than 10 years, male sex, poor diabetes control, neuropathy, foot deformity, high plantar foot pressures, and a history of amputation were significantly associated with the presence of foot ulceration. Meanwhile macrovascular and microvascular disease were not significant risk factors for ulceration (Lavery et al., 1998). On the other hand, Crowther et al. (2008) examined individuals with peripheral arterial disease-intermittent claudicatio and found higher levels of lower limb movement variability and a reduced walking speed compared to healthy age and mass matched controls.

5. Conclusion

In summary, our study demonstrates the presence of serious biomechanical alterations in the joints involved in both postural balance and gait not only in the sagittal plane but also in the coronal and transverse planes. Many of these alterations were detectable in diabetic patients without clinical evidence of neuropathy. The presence of both macro- and microangiopathies significantly aggravate this condition: thus, these findings have a remarkably important clinical fall out.

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References

Abbott, C.A., Vileikyte, L., Williamson, S., Carrington, A.L., Boulton, A.J., 1998. Multicenter study of the incidence of and predictive risk factor for diabetic neuropathic foot ulceration. Diabetes Care 21, 1071.

Altman, D.G., 1991. Practical Statistics for Medical Research. Chapman & Hall, London.

American Diabetes Association, 1988. American Academy of Neurology, consensus statement, report and recommendations of the San Antonio conference on diabetic neuropathy. Diabetes Care 11, 592–597.

Andersen, H., Gadeberg, P.C., Brock, B., Jakobsen, J., 1997. Muscular atrophy in diabetic neuropathy stereological magnetic resonance imaging study. Diabetologia 40, 1062.

Andersen, H., Jakobsen, J., 1999. Motor function in diabetes. Diabetes Rev. 7, 326.

Bevans, J., 1992. Biomechanics and plantar ulcers in diabetes. Foot 2, 166–172.

Bourdiol, J.R., 1980. Pied et statique. Maisonneuve, Paris. pp. 210–226.

Cappozzo, A., Catani, F., Della Croce, U., Leardini, A., 1995. Position and orientation in space of bones during movement, anatomical frame definition and determination. Clin. Biomech. 10, 171–178.

Cavanagh, P.R., Simoneau, G.G., Ulbrecht, J.S., 1993. Ulceration, unsteadiness, and uncertainty, the biomechanical consequences of diabetes mellitus. J. Biomech. 26 (1), 23–40.

Cavanagh, P.R., Ulbrecht, J.S., Caputo, G.M., 2000. New development in the biomechanics of the diabetic foot. Diabetes Metab. Res. Rev. 16 (1), S6–S10.

Courtemanche, R., Teasdale, N., Boucher, P., Fleury, M., Lajoie, Y., Bard, C., 1996. Gait problems in diabetic neuropathic patients. Arch. Phys. Med. Rehab. 77, 849–855.

Crowther, R.G., Spinks, W.L., Leicht, A.S., Quigley, F., Golledge, J., 2008. Lower limb movement variability in patients with peripheral arterial disease. Clin. Biomech. 23 (8), 1080–1085.

D'Ambrogi, E., Giurato, L., D'Agostino, M.A., Giacomozzi, C., Macellari, V., Caselli, A., Uccioli, L., 2003. Contribution of plantar fascia to the increased forefoot pressures in diabetic patients. Diabetes Care 26, 1525.

Della Croce, U., Leardini, A., Chiari, L., Cappozzo, A., 2005. Human movement analysis using stereophotogrammetry: Part 4: assessment of anatomical landmark misplacement and its effects on joint kinematics. Gait Posture 21 (2), 226–237.

Dickstein, R., Shupert, C.L., Horak, F.P., 2001. Fingertip touch improves postural stability in patients with peripheral neuropathy. Gait Posture 14 (3), 238–247.

Dingwell, J.B., Ulbrecht, J.S., Boch, J., Becker, M.B., O'Gorman, J.T., Cavanagh, P.R., 1999. Neuropathic gait shows only trends towards increased variability of sagittal plane kinematics during treadmill locomotion. Gait Posture 10 (1), 21–29.

Dingwell, J.B., Cusumano, J.P., Sternad, D., Cavanagh, P.R., 2000. Slower speeds in patients with diabetic neuropathy lead to improved local dynamic stability of continuous overground walking. J. Biomech. 33, 1269–1277.

Dingwell, J.B., Cavanagh, P.R., 2001. Increased variability of continuous overground walking in neuropathic patients is only indirectly related to sensory loss. Gait Posture (1), 1–10.

Donovan, J.L., Dequan, Z., Mueller, M.J., 2008. Pressure gradient and subsurface shear stress on the neuropathic forefoot. Clin. Biomech. 23 (3), 342–348.

Feldman, E.L., Stevens, M.J., Thomas, P.K., Browne, M.B., Canal, N., Greene, D.A., 1994. A practical two-step quantitative clinical and electrophysiological assessment for the diagnosis and staging of diabetic neuropathy. Diabetes Care 17 (11), 1281–1289.

Giacomozzi, C., Caselli, A., Macellari, V., Giurato, L., Lardieri, L., Uccioli, L., 2002. Walking strategy in diabetic patients with peripheral neuropathy. Diabetes Care 25, 1451.

Horak, F.B., Dickstein, R., Peterka, R.J., 2002. Diabetic neuropathy and surface sway-referencing disrupt somatosensory information for postural stability in stance. Somatosens. Mot. Res. 19 (4), 316–326.

Kapteyn, T.S., Njikoktjien, C.J., Bles, W., Kodde, L., Massen, C.H., Mol, J.M.F., 1983. Standardization in platform stabilometry being a part of posturography. Agressologie 24, 321–326.

Katoulis, E.C., Ebdon-Parry, M., Hollis, S., Harrison, A.J., Vilejkite, L., Kulkarni, J., et al., 1997. Postural instability in neuropathics diabetics patients at risk of foot ulceration. Diabetes Med. 14, 296–300.

Lavery, L.A., Armstrong, D.G., Vela, S.A., Quebedeaux, T.L., Fleischli, J.G., 1998. Practical criteria for screening patients at high risk for diabetic foot ulceration. Arch. Intern. Med. 158, 157.

Leardini, A., Sawacha, Z., Paolini, G., Nativo, R., Ingrosso, S., Benedetti, M.G., 2007. A new anatomical based protocol for gait analysis in children. Gait Posture 26, 560–571.

Maynard, V., Bakheit, A.M.O., Oldham, J., Freeman, J., 2003. Intra-rater and inter-rater reliability of gait measurements with CODA mpx30 motion analysis system. Gait Posture 17 (1), 59–67.

Mueller, M.J., Sinacore, D.R., Hoogstrate, S., Daly, L., 1994. Hip and ankle walking strategies, effect on peak plantar pressures and implications for neuropathic ulceration. Arch. Phys. Med. Rehab. 75 (11), 1196–1200.

Oomichi, T., Emoto, M., Tabata, T., Morioka, T., Tsujimoto, Y., Tahara, H., Shoji, T., Nishizawa, Y., 2006. Impact of glycemic control on survival of diabetic patients on chronic regular hemodialysis: a 7-year observational study. Diabetes Care 29 (7), 1496–1500.

Perkins, BA., Bril, V., 2003. Diabetic neuropathy: a review emphasizing diagnostic methods. Clin. Neurophysiol. 114 (7), 1167–1175.

Perry, J., 1992. Gait Analysis, Normal and Pathological Function. McGraw-Hill, New York. pp. 435–436.

Sawacha, Z., Cristoferi, G., Guarneri, G., Corazza, S., Don, G., Facchinetti, A., Denard, C., Sommavilla, M., Zaccaria, M., Avogaro, A., Cobelli, C., 2005. A method for the simultaneous assessment of gait and posture. In: Proceedings Siamoc 2005, Pisa, Italia.

Sawacha, Z., Corazza, S., Donà, G., Denti, P., Facchinetti, A., 2009. Characterizing multisegment foot kinematics during gait in diabetic foot patients J. NeuroEng. Rehab., in press.

Shaw, J.E., Van Schie, C.H.M., Carrington, A.L., Abbott, C.A., Boulton, A.J.M., 1998. An analysis of dynamic forces transmitted through the foot in diabetic neuropathy. Diabetes Care 21 (11), 1955–1959.

Simoneau, G.G., Ulbrecht, J.S., Derr, J.A., Becker, M.B., Cavanagh, P.R., 1995. Postural instability in patients with diabetic sensory neuropathy. Diabetes Care 17 (12), 1411–1421.

Sinacore, D.R., Bohnert, K.L., Hastings, M.K., Johnson, J.E., 2008. Mid foot kinetics characterize structural polymorphism in diabetic foot disease. Clin. Biomech. 23 (5), 653–661.

Uccioli, L., Caselli, A., Giacomozzi, C., Macellari, V., Giurato, L., Lardieri, L., Menzinger, G., 2001. Pattern of abnormal tangential forces in the diabetic neuropathic foot. Clin. Biomech. 16 (5), 445–446.

Van Schie, C.H.M., 2005. A Review of the Biomechanics of the Diabetic Foot. Int. J. Low Extrem. Wounds 4 (3), 160–170.